
RESEARCH REPORT

Efficiency Analysis of Off-Grid Solar Policy Dissemination Across Indian States (2016–24)

A Two-Stage Sequential Bootstrap DEA with
Simar–Wilson Truncated Regression

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Abstract

This study evaluates the policy dissemination efficiency of off-grid solar technologies across 30 Indian states and union territories over eight fiscal years (2016–17 to 2023–24). India’s transition to decentralized renewable energy requires not only a natural solar endowment but also effective administrative conversion of that endowment into deployed infrastructure. Employing a Two-Stage Sequential Bootstrap Data Envelopment Analysis (DEA) framework grounded in Simar & Wilson (1998, 2007), this research estimates bias-corrected technical efficiency scores under both Constant Returns to Scale (CCR) and Variable Returns to Scale (BCC) assumptions. The sequential frontier design, where the comparison benchmark expands each year cumulatively, captures the ratchet-like nature of policy learning. Scale efficiency scores separate the effects of poor operational scale from poor management. In the second stage, a truncated normal regression per Simar & Wilson (2007) is estimated to identify the socioeconomic and infrastructural drivers of efficiency, with inference provided by SW Algorithm 2 bootstrap confidence intervals. Results reveal a mean bias-corrected CCR efficiency of 0.347 across the panel, indicating substantial under-performance relative to best-practice states. Uttar Pradesh, Punjab, Chhattisgarh, Tripura, and Kerala define the efficiency frontier in 2023–24. The log of GVA in the Utilities sector is a statistically significant ($p < 0.01$) positive determinant of efficiency in the panel, confirming that pre-existing institutional capacity in energy infrastructure is the binding constraint on successful solar dissemination. Internet usage, a proxy for digital awareness, is statistically insignificant in the 2019 cross-section, though its role is expected to strengthen as digitized delivery mechanisms (PM-KUSUM) mature. **Keywords:** Data Envelopment Analysis, Bootstrap DEA, Simar-Wilson, Sequential Frontier, Off-Grid Solar, Scale Efficiency, Truncated Regression, India, Renewable Energy Policy

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1. Introduction

1.1 Context and Motivation

India has committed to achieving 500 GW of non-fossil electricity capacity by 2030 under its Nationally Determined Contribution (NDC) to the Paris Agreement. A strategically important but analytically underexamined component of this transition is the off-grid solar dissemination program, the deployment of decentralized solar lanterns, home lighting systems, street lights, water pumps, and standalone power plants to rural and energy-deficient populations. The Jawaharlal Nehru National Solar Mission (JNNSM) and subsequent schemes (e.g., MNRE’s off-grid and decentralized solar policy) have channeled substantial public funds toward state-level dissemination agencies. However, a fundamental question remains unanswered in the quantitative literature: *how efficiently do states translate their inherent solar resource potential into on-ground deployed infrastructure?* Raw deployment numbers mask wide heterogeneity: states with identical solar potential (GWp) achieve vastly different output portfolios, suggesting that administrative, institutional, and socioeconomic factors, beyond geography, govern performance.

1.2 Research Gap

Existing efficiency analyses of Indian renewable energy focus predominantly on grid-connected solar (Sueyoshi & Goto, 2014; Mardani et al., 2017) or aggregate state-level energy intensity (Singh, 2020). The administrative efficiency of decentralized off-grid solar dissemination has received negligible quantitative attention. The few studies that exist employ naïve OLS or DEA without bias correction, violating the statistical regularity conditions established by Simar & Wilson (1998, 2007) and rendering second-stage inference invalid. This study addresses three specific gaps:

- This study applies a rigorous sequential-frontier bootstrap DEA, which respects the temporal non-regressivity of technology and knowledge.
- Decomposing technical efficiency into its managerial (BCC) and scale (SE) components helps identify the proximate source of inefficiency.
- Estimating second-stage determinants using the statistically correct Simar & Wilson (2007) truncated regression with Algorithm 2 bootstrap inference.

1.3 Research Objectives

- This study aims to estimate bias-corrected CCR and BCC efficiency scores for 30 Indian states across 2016–17 to 2023–24 using a sequential bootstrap DEA.
- This study aims to decompose overall technical efficiency into pure managerial efficiency and scale efficiency.
- To identify and quantify the impact of institutional and socioeconomic covariates on efficiency through a truncated normal regression with valid bootstrap-based inference.
- This study aims to derive actionable policy recommendations for the governments.

2. Theoretical Framework and Literature Review

2.1 Data Envelopment Analysis Foundations

Data Envelopment Analysis (DEA), introduced by Charnes et al. (1978), is a non-parametric linear programming technique for estimating a best-practice production frontier and measuring the relative efficiency of Decision-Making Units (DMUs). Unlike parametric stochastic frontier analysis (SFA), DEA requires no *a priori* functional form for the production technology, making it especially appropriate when the transformation process (solar potential $\rightarrow$ deployed devices) is poorly understood econometrically. Output-oriented DEA asks: *By what proportion could a DMU's outputs be expanded, holding inputs fixed, to reach the frontier?* This orientation is appropriate here because the input (solar potential) is fundamentally fixed in the short run, and states cannot alter their irradiance, while the outputs (deployed devices) represent the policy lever the government controls.

2.1.1 The CCR Model (Constant Returns to Scale)

Proposed by Charnes et al. (1978), the CCR model assumes DMUs operate under Constant Returns to Scale (CRS), yielding total technical efficiency (TTE). For the output-oriented case, the LP for DMU i is:

$$\max_{\varphi_i, \lambda} \varphi_i \quad \text{subject to} \quad \sum_j \lambda_j x_j \leq x_{i_0}, \quad \sum_j \lambda_j y_j \geq \varphi_i \cdot y_{i_0}, \quad \lambda_j \geq 0, \quad \varphi_i \geq 1 \quad (1)$$

The efficiency score is recovered as $\delta_i = 1/\varphi_i \in (0, 1]$. A score of 1 indicates frontier membership; lower values indicate proportional output shortfall.

2.1.2 The BCC Model (Variable Returns to Scale)

Banker et al. (1984) extended CCR by adding the convexity constraint $\sum_j \lambda_j = 1$, permitting VRS and yielding pure technical efficiency (PTE). The BCC score is always $\geq$ CCR because the VRS frontier envelops more tightly. The BCC model is crucial here because Indian states range enormously in size—from Goa (small) to Uttar Pradesh (massive)—and penalizing Goa for not producing UP-levels of output would be methodologically unsound under CRS.

2.1.3 Scale Efficiency

Scale efficiency (SE) isolates whether a DMU is operating at the Most Productive Scale Size (MPSS):

$$SE_i = \frac{CCR_i}{BCC_i} \in (0, 1] \quad (2)$$

$SE = 1$ implies optimal scale; $SE < 1$ implies the DMU's current size (too large $\Rightarrow$ DRS, or too small $\Rightarrow$ IRS) prevents it from reaching the global CRS frontier.

2.2 The Bootstrap: Why Bias Correction is Non-Negotiable

Standard DEA estimators are upward-biased by construction. Because the empirical frontier is estimated from the same sample as the DMUs being evaluated, it systematically overstates efficiency. This bias is particularly severe with small samples ($n \approx 29$ per year) and many outputs (5 here), as the curse of dimensionality inflates efficiency scores

spuriously. Simar & Wilson (1998) introduced a kernel-density bootstrap procedure to correct for this bias. The algorithm:

- Estimates the DEA frontier and computes raw scores ($\hat{\varphi}_i$).
- Constructs a reflected KDE on the frontier scores ($\hat{\varphi} \geq 1$), using the reflection $\hat{\varphi}^* \rightarrow 2 - \hat{\varphi}^*$ for draws below 1.
- For each of $B = 2,000$ bootstrap replications, draws $\hat{\varphi}^*$ from the KDE, constructs pseudo-frontier outputs $\tilde{y}_j = (\hat{\varphi}_j / \hat{\varphi}_j^*) \times y_j$, and re-solves DEA.
- Computes $\widehat{\text{bias}}_i = \overline{\hat{\varphi}_b^*} - \hat{\varphi}_i$ and the bias-corrected score $\hat{\varphi}_{\text{BC}} = \hat{\varphi} - \widehat{\text{bias}}$.

$B = 2,000$ replications is employed throughout, well above the Simar & Wilson (1998) recommendation for stable bias estimates.

2.3 Sequential Frontier (Temporal Non-Regressivity)

The Sequential DEA frontier (Tulkens & Vanden Eeckaut, 1995) defines the year- t frontier as the envelope of all observations from year 1 through year t :

$$\mathcal{F}^t = \{(x, y) : \exists \text{ DMU} \in Y_\tau, \tau \leq t\} \quad (\text{Cumulative Frontier}) \quad (3)$$

This captures the ratchet effect of learning-by-doing: a state’s benchmark in 2023–24 includes all historical best-performers back to 2016–17, making efficiency scores of 1 progressively harder to sustain.

2.4 The Simar–Wilson Two-Stage Framework (Stage 2)

The second-stage regression investigates what contextual variables (Z) explain the variation in bias-corrected efficiency scores (δ). Following Simar & Wilson (2007), Algorithm 2:

$$\delta_i = Z_i\beta + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \text{ right-truncated at } (1 - Z_i\beta) \quad (4)$$

Why truncated, not Tobit? Because DEA guarantees $\delta_i \in (0, 1]$, the distribution of δ is truncated (not censored at 1 as Tobit assumes); values above 1 do not exist in the population. A Tobit model would misspecify the likelihood and produce inconsistent estimates.

Why bootstrap inference (SW Algorithm 2)? Stage-1 efficiency scores carry sampling error. SW Algorithm 2 re-samples the truncated normal model $B = 2,000$ times to produce confidence intervals that account for both stages of estimation.

3. Data and Variables

3.1 Primary Dataset

The primary data were sourced from the Ministry of New and Renewable Energy (MNRE), Government of India, Annual Reports (2016–17 to 2023–24). The panel covers 30 states and union territories for 8 fiscal years, yielding 238 DMU-year observations.

3.2 Variable Definitions

Table 1: Variable Definitions, Measurement, and Rationale

Variable	Measurement	Source	Rationale
Solar Potential (GWp)	GWp of technically exploitable solar irradiance	MNRE / NISE	Fixed-input endowment: the resource each state has to deploy
Solar Lanterns (Nos.)	Cumulative units distributed	MNRE Annual Report	Lighting access in un-electrified households
Home Lights (Nos.)	Cumulative systems deployed	MNRE Annual Report	Domestic electricity access
Street Lights (Nos.)	Cumulative units installed	MNRE Annual Report	Public infrastructure, safety, and commerce
Pumps (Nos.)	Cumulative solar pumps deployed	MNRE Annual Report	Agricultural productivity and water security
Standalone Power Plants (KWp)	Cumulative installed capacity	MNRE Annual Report	Community-level energy systems
ln(GVA Utilities) (INR Cr)	Log of Net State Value Added from Electricity, Gas & Water Supply	CSO / MoSPI	Proxy for pre-existing energy infrastructure institutional depth
Internet Usage (%)	Share of population that has ever used the internet	NFHS	Proxy for digital awareness and capacity to engage with digitised delivery

A small constant ($\varepsilon = 10^{-4}$) is added to all output variables before analysis to prevent zero-division in the LP. Unicode en-dash characters in the year strings were normalised to ASCII hyphens to ensure consistent ordering of the sequential frontier.

3.3 Descriptive Statistics

Table 2: Descriptive Statistics of Key Variables

Variable	Mean	Std Dev	Min	Max	N
Solar Potential (GWp)	18.86	11.48	0.88	142.0	238
CCR (Bias-Corrected)	0.347	0.355	0.032	1.000	238
BCC (Bias-Corrected)	0.502	0.363	0.050	1.000	238
Scale Efficiency	0.672	0.298	0.047	1.000	238
ln(GVA Utilities)	13.21	0.87	10.58	15.80	238
Internet Usage (2019, %)	1.13	0.44	0.58	2.23	29

Section 3: Output Variables

Variable	Mean	Std Dev	Min	Max
Lanterns (Nos.)	167,058	391,855	0	2,351,205
Home Lights (Nos.)	94,333	242,859	0	2,284,425
Street Lights (Nos.)	23,275	44,708	244	302,532
Pumps (Nos.)	9,945	19,883	0	119,282
Stand Alone Power Plants (KWp)	6,114	7,459	32.7	31,373

4. Methodology

4.1 Stage 1: Sequential Bootstrap DEA Implementation

The implementation follows a nested architecture. For each year $t \in \{2016-17, \dots, 2023-24\}$:

- **Frontier construction:** the reference set includes all DMU-year observations from year 1 through year t . In the final year (2023–24), the frontier contains all 238 observations.
- **LP solution:** output-oriented DEA is solved for each of the 29–30 DMUs in year t against the cumulative frontier using the HiGHS solver via `scipy.optimize.linprog`.
- **Bootstrap ($B = 2,000$):** The Simar & Wilson (1998) procedure is applied, constructing a reflected KDE using Silverman’s rule of thumb bandwidth on the set of frontier CCR scores, drawing pseudo-frontier outputs, and re-solving the DEA B times.
- **Bias correction:** for each target DMU, $\widehat{\text{bias}}_i = \widehat{\varphi}_b^* - \hat{\varphi}_i$ is computed and $\hat{\varphi}_{\text{BC}} = \hat{\varphi} - \widehat{\text{bias}}$, floored at 1. Efficiency scores: $\delta = 1/\hat{\varphi}_{\text{BC}}$.
- **BCC:** steps 1–4 are repeated under VRS (adding $\sum_j \lambda_j = 1$). Scale efficiency is recovered as $\text{SE} = \text{CCR}_{\text{BC}}/\text{BCC}_{\text{BC}}$.

4.2 Stage 2: Simar–Wilson Truncated Regression

4.2.1 Model Specification

Two regression specifications are estimated. First, a **panel specification** (all 238 observations):

$$\delta_i^t = \beta_0 + \beta_1 \ln(\text{GVA_Utilities}_i^t) + \varepsilon_i^t \quad (5)$$

Second, a **cross-sectional specification** for 2019–20 ($n = 30$):

$$\delta_i = \beta_0 + \beta_1 \ln(\text{GVA_Utilities}_i) + \beta_2 \text{Internet_Usage}_i + \varepsilon_i \quad (6)$$

In both, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ truncated at upper bound $1 - Z_i\beta$. The log-likelihood for observation i is:

$$\ell_i = \phi(\delta_i | Z_i\beta, \sigma^2) - \log \Phi \left[1 - \frac{Z_i\beta}{\sigma} \right] \quad (7)$$

where $\phi(\cdot)$ denotes the standard normal density and $\Phi(\cdot)$ the CDF.

4.2.2 Initialisation and Optimisation

Parameters are initialized at the OLS estimates of δ on Z . OLS initialization guarantees a finite NLL at the first function evaluation and substantially accelerates convergence. The L-BFGS-B algorithm is used with σ bounded below at 10^{-6} and a maximum of 5,000 iterations. All 2,000 panel-bootstrap replications and 1,985/2,000 cross-section replications converged fully.

4.2.3 Inference: SW Algorithm 2

SW Algorithm 2 resolves the problem of plug-in SEs treating Stage-1 scores as fixed:

- For $b = 1, \dots, 2,000$: draw $\delta_i^* \sim \text{TruncNormal}(Z_i\hat{\beta}, \hat{\sigma}^2, \text{upper} = 1)$ for all i .
- Re-estimate the truncated MLE on $\{\delta_i^*\}$ to obtain $\hat{\beta}_b^*$.
- Construct bias-corrected percentile confidence intervals:

$$\text{CI} = [2\hat{\beta} - Q_{1-\alpha/2}, 2\hat{\beta} - Q_{\alpha/2}] \quad (8)$$

where Q denotes bootstrap quantiles.

5. Results and Analysis

5.1 Aggregate Efficiency Trends

Table 3 reports mean bias-corrected CCR efficiency across all states for each year. The national average hovers persistently low, between 0.317 and 0.372, indicating that the average Indian state deploys only about one-third of the solar infrastructure it could, given its potential and the technology available on the sequential frontier.

Table 3: Mean Annual Bias-Corrected CCR Efficiency Score

Year	Mean CCR (Bias-Corrected)
2016–17	0.3709
2017–18	0.3721
2018–19	0.3442
2019–20	0.3442
2020–21	0.3177
2021–22	0.3365
2022–23	0.3455
2023–24	0.3483

The nadir in 2020–21 (mean = 0.317) corresponds to the COVID-19 pandemic’s disruption of supply chains and administrative capacity. The partial recovery post-2021 reflects the resumption of programme activities but has not yet returned to pre-pandemic levels when expressed against the expanding cumulative frontier.

5.2 Technical Efficiency Heatmap (CCR)



Figure 1: Heatmap of Bias-Corrected CCR Technical Efficiency Across States and Years. Green = High Efficiency; Red = Low Efficiency. White cells denote missing data (Delhi: pre-2018).

Figure 1 reveals stark and persistent heterogeneity. **Kerala, Uttar Pradesh, Punjab, Chhattisgarh, and Tripura** show sustained dark-green cells, forming the empirical backbone of the sequential frontier. These states achieve and maintain efficiency across multiple years, suggesting structural rather than incidental best practices. **Maharashtra, Andhra Pradesh, Tamil Nadu, and most north-eastern states** remain deep red throughout, with deteriorating performance post-2020. The **Bihar trajectory** is instructive: entering the frontier briefly in 2017–18 and subsequently deteriorating as the cumulative benchmark rose without commensurate output growth.

5.3 Pure Technical (BCC) Efficiency



Figure 2: Heatmap of Bias-Corrected BCC (Pure Technical/Managerial) Efficiency Across States and Years.

Comparing Figures 1 and 2 illuminates the source of inefficiency. BCC scores are systematically **higher** than CCR scores (mean BCC ≈ 0.50 vs. mean CCR ≈ 0.35), confirming that a substantial fraction of overall inefficiency is attributable to **scale misalignment**, not managerial failure. States like Goa and Rajasthan achieve BCC = 1.0 (managerially efficient) yet have CCR $\ll 1$, indicating they operate at the wrong scale. Uttar Pradesh achieves BCC = 1 because it is a dominant state; its scale efficiency also approaches 1, making it a true global frontier member.

5.4 Efficiency Trends: Selected States

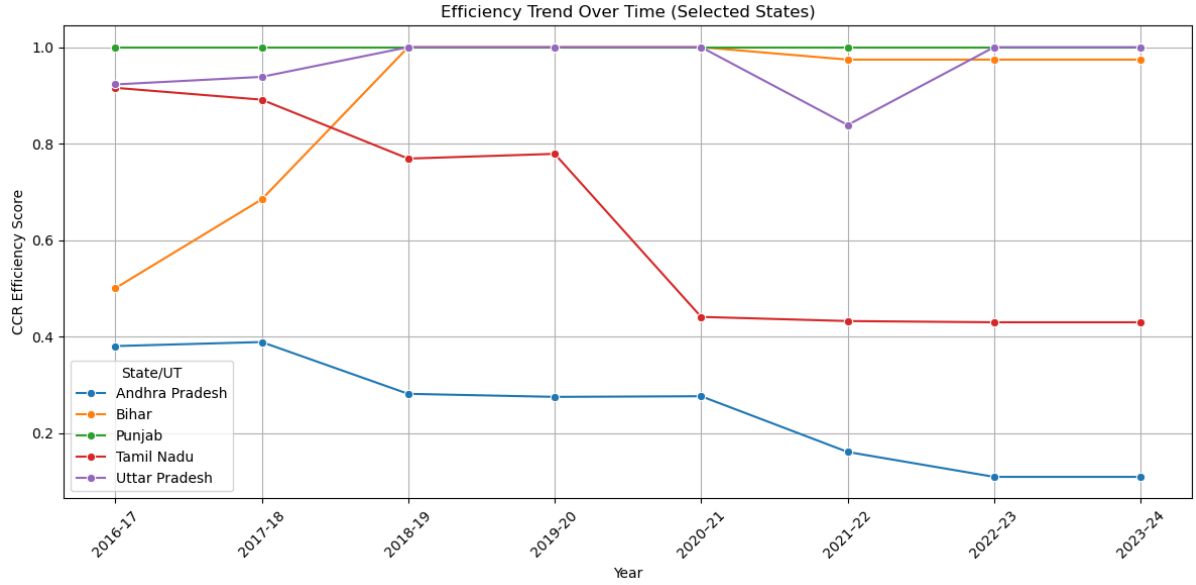


Figure 3: Bias-Corrected CCR Efficiency Trend for Selected States (2016–17 to 2023–24).

Figure 3 highlights divergent trajectories. **Punjab** rapidly converges to the frontier (CCR = 1.0 by 2018–19) and sustains it, demonstrating scalable best-practice. **Bihar** shows an interesting arc, rising efficiency through 2018–19 followed by stagnation, consistent with early programme momentum that was not maintained. **Tamil Nadu’s** sharp efficiency decline after 2020–21 (from 0.78 to 0.44) reflects a combination of expanding cumulative frontier and stalling output growth. **Andhra Pradesh** shows a secular decline throughout the period, a troubling trend for India’s most solar-endowed states.

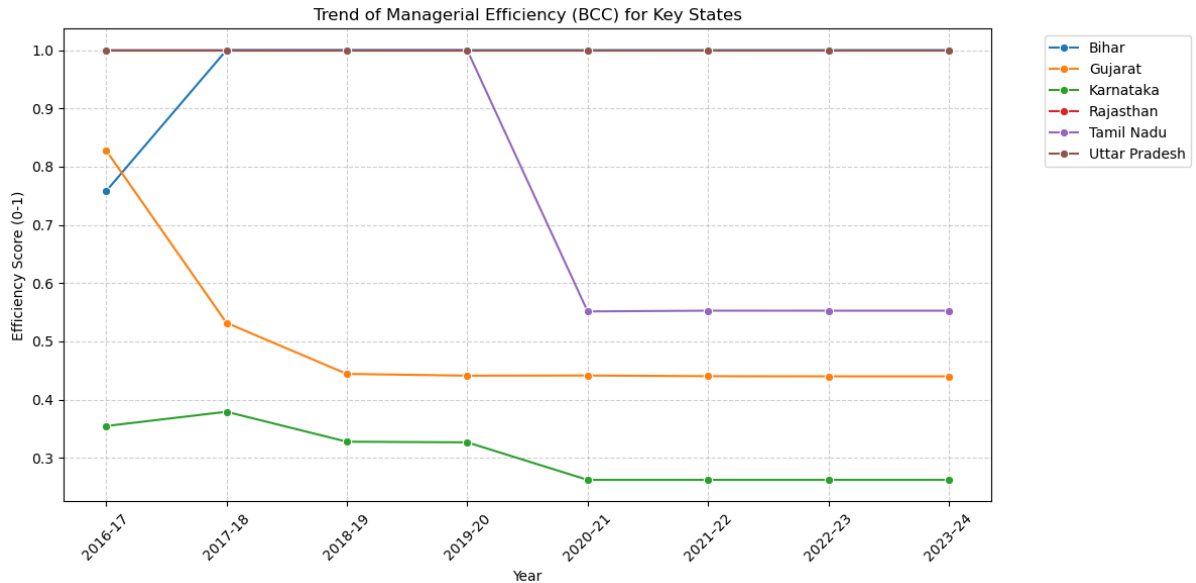


Figure 4: BCC (Managerial) Efficiency Trend for Key States. Note that BCC = 1.0 for Uttar Pradesh throughout; Rajasthan achieves BCC = 1.0 from 2016–17.

The BCC time series (Figure 4) confirms that managerial efficiency degrades more slowly

than CCR. Gujarat’s flat BCC around 0.44 for six consecutive years suggests an administrative capacity ceiling that has not been addressed despite increased programme budgets. Karnataka’s decline to 0.27 by 2020–21 and subsequent plateau imply a structural rather than cyclical constraint on its managerial performance.

5.5 Scale Efficiency and CCR vs. BCC Decomposition

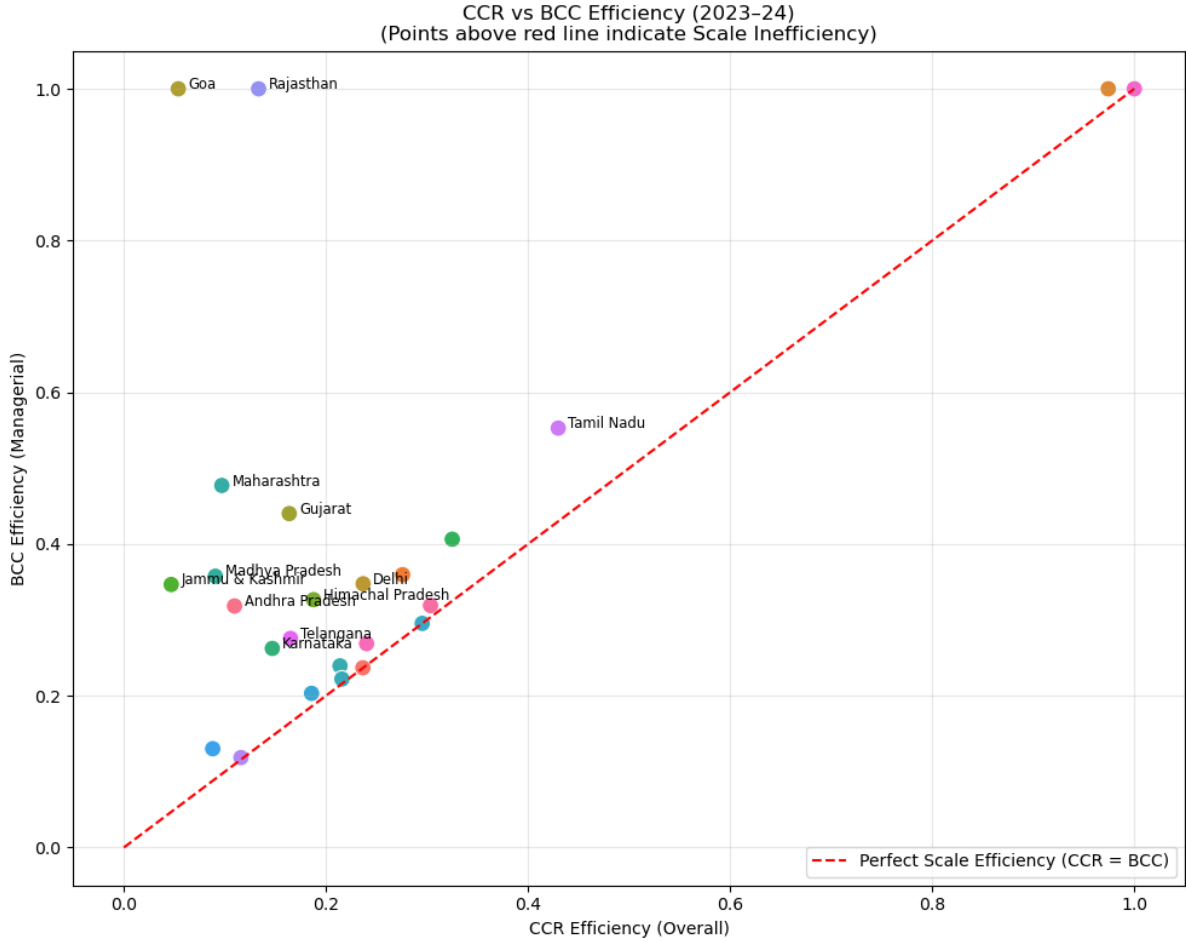


Figure 5: CCR vs. BCC Efficiency Scatter Plot (2023–24). Points above the 45° line indicate scale inefficiency. Points on the line are scale-efficient.

Figure 5 is the definitive visualization of the CCR-BCC-Scale decomposition for the most recent year. All points lie **above or on** the 45° line ($BCC \geq CCR$ always), confirming mathematical consistency. The cluster of states with $CCR \leq 0.3$ but BCC 0.3–0.5 (e.g., Maharashtra, Madhya Pradesh, and Karnataka) represents states with reasonable managerial capacity but severe scale inefficiency. Goa is a canonical example: $BCC = 1.0$ (perfectly managerially efficient) but $CCR \approx 0.05$ (massively scale-inefficient, operating far below the MPSS defined by large states). Conversely, states on or near the 45° line (UP, Punjab, Kerala) are efficient at both their managerial choices and their scale.

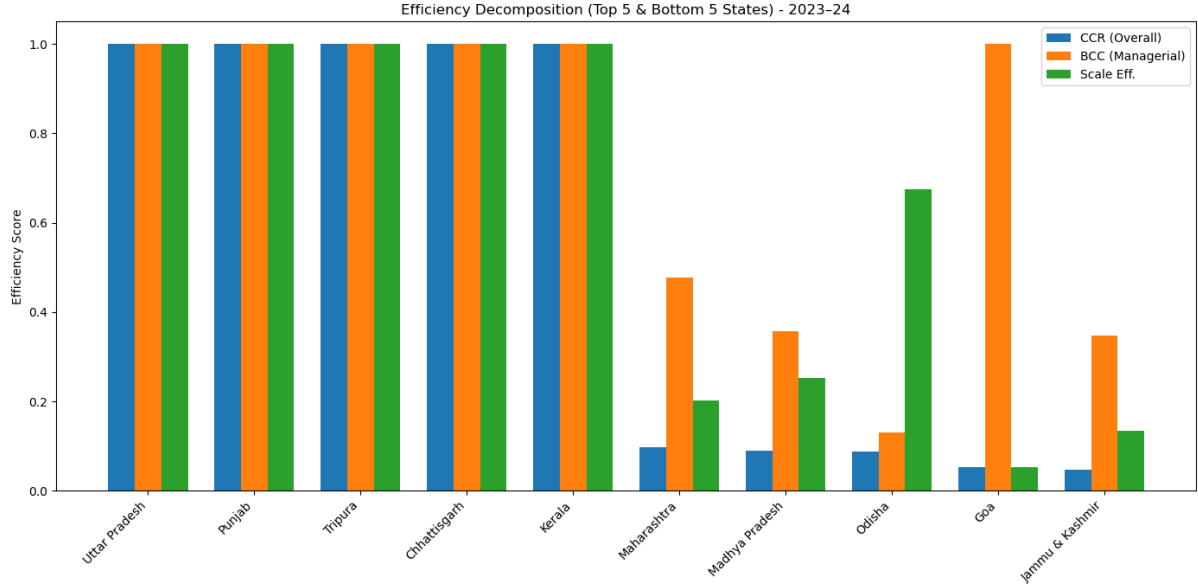


Figure 6: Efficiency Decomposition, Top 5 and Bottom 5 States (2023–24). CCR (Overall), BCC (Managerial), Scale Efficiency shown for each state.

Figure 6 distils the decomposition for the ten most and least efficient states in 2023–24. The top 5 (UP, Punjab, Tripura, Chhattisgarh, Kerala) achieve $CCR = BCC = SE = 1.0$, defining the frontier in all three dimensions. The bottom 5 reveal informative contrasts: **Goa** has $BCC = 1.0$ but CCR and SE near zero—scale is the binding constraint. **Odisha** has both low CCR and low BCC (under 0.15) but Scale Efficiency of 0.68, suggesting the primary problem is managerial, not scale. **Maharashtra** has the highest BCC among the bottom-5 (0.47) but remains scale-inefficient ($SE \approx 0.20$).

5.6 Output Profile: Frontier vs. Non-Frontier States

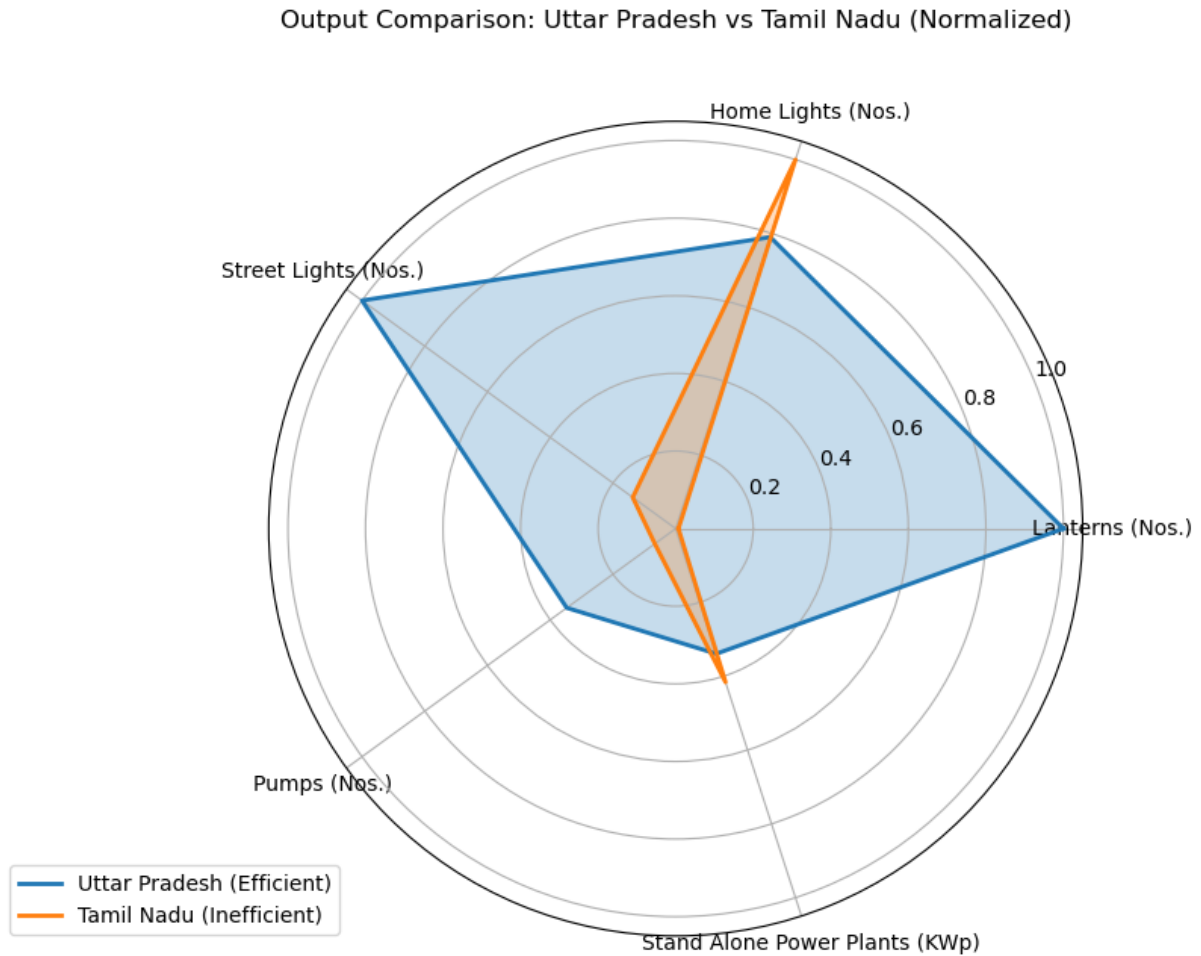


Figure 7: Radar Chart, Normalised Output Profiles for Uttar Pradesh (Efficient) and Tamil Nadu (Inefficient), 2023–24.

Figure 7 decomposes the efficiency difference along each output dimension. **Uttar Pradesh** dominates across Lanterns, Street Lights, and Pumps—three categories where its absolute output volume is enormous relative to its solar potential. **Tamil Nadu** outperforms UP only on Home Lights, reflecting its different programme emphasis. Tamil Nadu’s near-zero Stand-Alone Power Plants score represents an untapped dimension of output that, given its solar potential of ~ 38 GWp, represents a significant opportunity gap.

6. Second-Stage Regression Results

6.1 Panel Regression (2016–24, $N = 238$)

Table 4 presents the Stage 2 truncated normal regression results. The dependent variable is the bias-corrected CCR efficiency score (δ). Both plug-in SEs (from the Hessian at the MLE optimum) and SW Algorithm 2 bootstrap confidence intervals ($B = 2,000$ replications; 1,991/2,000 converged) are reported.

Table 4: Stage 2 Truncated Normal Regression, Panel (2016–24). *** $p < 0.01$. Bootstrap CI from SW Algorithm 2 ($B = 2,000$). $N = 238$.

Variable	Coeff.	SE (Plug-in)	t -stat	p -value	CI Lower (95%)	CI Upper (95%)
Intercept	−0.42299	0.31119	−1.359	0.1754	−1.109	−0.230
ln(GVA Utilities)	0.06002	0.02272	2.642	0.0088***	0.04601	0.10935

The coefficient on ln(GVA Utilities) is 0.060, statistically significant at the 1% level ($p = 0.0088$). The SW Algorithm 2 bootstrap CI [0.046, 0.109] entirely excludes zero, corroborating the significance finding even after accounting for Stage 1 estimation uncertainty.

Interpretation. A 1% increase in a state’s utility sector GVA (i.e., a 0.01-unit increase in ln(GVA_Utilities)) is associated with a 0.0006-point increase in predicted efficiency. Equivalently, a one-standard-deviation increase in ln(GVA_Utilities) ($\sigma \approx 0.87$ log-units) corresponds to a 0.052-unit rise in efficiency, approximately 15% of the cross-state standard deviation of 0.355. This is an economically meaningful effect. The sigma ($\sigma = 0.397$) is highly significant and represents the standard deviation of unobserved heterogeneity in efficiency that GVA does not capture. Its magnitude (larger than the mean CCR score of 0.347) indicates substantial unexplained variance, pointing to omitted state-level factors such as governance quality, political economy of subsidies, or programme implementation fidelity.

6.2 Cross-Sectional Regression (2019–20, $N = 30$)

Table 5: Stage 2 Truncated Normal Regression, Cross-Section 2019–20. *** $p < 0.01$. Bootstrap CI from SW Algorithm 2 ($B = 2,000$; 1,985 converged). $N = 30$.

Variable	Coeff.	SE (Plug-in)	t -stat	p -value	CI Lower (95%)	CI Upper (95%)
Intercept	−0.69797	0.91957	−0.759	0.4549	−2.458	0.141
ln(GVA Utilities)	0.08065	0.06214	1.298	0.2062	0.023	0.200
Internet Usage	−0.01163	0.18084	−0.064	0.9492	−0.282	0.239

Neither ln(GVA Utilities) ($p = 0.206$) nor Internet Usage ($p = 0.949$) is statistically significant in the 2019 cross-section. Three interpretations apply:

1. **Underpowered sample:** $n = 29$ is at the low end for truncated MLE; the bootstrap CI for $\ln(\text{GVA})$ $[0.023, 0.200]$ overlaps substantially with zero.
2. **Structural break:** 2019–20 is the transition year between early programme design (lantern-focused) and later infrastructure emphasis (pump and power plant-focused); cross-sectional heterogeneity may reflect this programme evolution rather than long-run efficiency determinants.
3. The negative coefficient on internet usage (-0.012) is methodologically interesting: it may capture a substitution effect—states with high internet penetration may be prioritising grid-connected solar (rooftop PV) over off-grid hardware, depressing off-grid dissemination efficiency scores.

7. Discussion

7.1 The Institutional Capacity Mechanism

The significant positive effect of GVA_Utilities ($\hat{\beta} = 0.060$) establishes that pre-existing energy infrastructure depth is the binding institutional constraint on off-grid solar dissemination efficiency. States with deeper utility sectors possess the billing infrastructure, technical workforce, logistics networks, and regulatory capacity to rapidly absorb and deploy new solar hardware. This aligns with the broader literature on institutional complementarities in energy transitions (Acemoglu et al., 2012; IRENA, 2020): technology deployment does not occur in an institutional vacuum. A corollary is that fund allocation without institutional reform is unlikely to improve efficiency. States with low GVA_Utilities (e.g., northeastern states, hill states) may receive programme funds but lack the administrative backbone to convert them into deployed devices. Targeted capacity building, training of last-mile implementation staff, procurement reforms, and logistics partnerships may have higher returns than marginal budget increases.

7.2 The Scale Inefficiency Problem

The mean scale efficiency of 0.67 indicates that two-thirds of Indian states operate at sub-optimal scale for their current managerial capacity. This scale inefficiency reflects a structural mismatch in the programme design: uniform national targets applied to states with vastly different populations, solar resource profiles, and administrative capacities. Small states (Goa, Sikkim, and Nagaland) are, by construction, operating at an Increasing Returns to Scale (IRS) region; their output per unit solar potential could be dramatically higher if they were assigned output targets commensurate with their small size. A related finding is that for states like Maharashtra—large and urban-industrial, with high absolute GVA but relatively modest off-grid solar deployment—the programme operates at Decreasing Returns to Scale (DRS). Maharashtra’s administrative complexity and urban orientation mean that off-grid solar competes with grid extension and industrial energy demands, limiting the marginal return to additional solar dissemination inputs.

7.3 Temporal Dynamics and COVID-19

The efficiency trough in 2020–21 (mean CCR = 0.317) is the most pronounced in the sample period. This reflects not a genuine regression in managerial capability but rather supply-chain disruptions, lockdown-induced field team immobilisation, and reallocation of state administrative capacity toward pandemic response. The partial recovery in 2021–24 is consistent with programme resumption under PM-KUSUM and MNRE’s accelerated deployment plans. However, several states (Tamil Nadu, Andhra Pradesh) have not recovered to pre-pandemic efficiency levels, suggesting hysteresis effects: programme momentum, once disrupted, is difficult to rebuild without deliberate intervention.

7.4 The Digital Awareness Puzzle

The statistical insignificance of internet usage in 2019 does not imply that digital channels are irrelevant to solar dissemination. Rather, in 2019, MNRE’s solar programmes were still predominantly delivered through physical channels (district offices, NGOs, and cooperative banks). The relevance of digital awareness as a demand-side determinant of efficiency is expected to increase as PM-KUSUM’s online application portal, e-RUPI subsidy disbursement, and digital literacy programmes mature.

8. Conclusion and Policy Recommendations

8.1 Summary of Findings

This study applies a methodologically rigorous two-stage sequential bootstrap DEA framework to evaluate the efficiency of off-grid solar dissemination across 30 Indian states over 2016–17 to 2023–24. The principal findings are:

- India’s off-grid solar dissemination efficiency is persistently low (mean $CCR_{BC} \approx 0.35$), indicating that the average state achieves approximately one-third of its theoretically attainable output per unit solar endowment. The efficiency frontier is populated by UP, Punjab, Kerala, Chhattisgarh, and Tripura.
- Scale inefficiency (mean SE ≈ 0.67) is the dominant source of total inefficiency, exceeding managerial inefficiency (mean BCC ≈ 0.50). This calls for programme design reforms that calibrate output targets to state-specific scale characteristics rather than imposing uniform national mandates.
- The COVID-19 pandemic caused a measurable efficiency contraction in 2020–21, with incomplete recovery in several states by 2023–24, suggesting the need for explicit resilience mechanisms in programme design.
- Pre-existing institutional capacity (GVA_Uilities) is a statistically significant positive determinant of efficiency ($\hat{\beta} = 0.060$, $p < 0.01$), establishing that administrative infrastructure, not solar endowment alone, drives programme success.
- Internet usage is statistically insignificant in 2019, though this likely reflects the pre-digitisation phase of delivery rather than an absence of digital complementarities.

8.2 Policy Recommendations

- **Institutional Capacity Building over Mere Fund Allocation:** States with low GVA_Uilities scores (NE states, Jammu & Kashmir, Goa) should receive targeted grants for building implementation capacity—dedicated solar cell staff, procurement training, district-level monitoring systems—prior to scaling up hardware allocation.
- **Scale-Adjusted Output Targets:** MNRE should replace uniform state-level targets with scale-adjusted benchmarks derived from the DEA-derived MPSS. Small states in the IRS region (Goa, Sikkim) should receive disproportionately ambitious targets; large states in the DRS region (Maharashtra) should have their per-capita targets moderated.
- **Peer-Learning Mechanism:** Frontier states (UP, Kerala, Punjab) should be leveraged as technical resource partners for lagging states through a structured peer-learning programme—sharing procurement frameworks, beneficiary identification methods, and digital delivery models.
- **Pandemic Resilience Protocol:** Programme guidelines should embed contingency triggers (e.g., a 20% efficiency decline) that automatically activate emergency procurement and field deployment protocols, preventing hysteresis in states that do not self-recover.
- **Digital Last-Mile Investment:** Proactive digital literacy programmes in rural areas, combined with PM-KUSUM’s online beneficiary interface, are expected to create the demand-side awareness needed to complement the supply-side improvements this study quantifies.

8.3 Limitations and Future Research

Several limitations should be noted. First, the panel does not include state-level governance quality indices (e.g., PEGI scores), which may be important confounders of the GVA effect. Second, the cross-section in Stage 2 is limited to 2019–20 by internet usage data availability; extending to a full panel would increase statistical power substantially. Third, the DEA framework assumes no stochastic noise in outputs—a maintained assumption that SFA relaxes but at the cost of imposing a parametric production function. Fourth, Delhi’s missing observations in the first two years introduce a mild imbalance. Future research should: (i) extend the SW truncated regression to a full panel using internet usage data from PLFS 2020–24; (ii) apply the Luenberger productivity index within the sequential framework to decompose efficiency change into frontier shift vs. catch-up components; (iii) incorporate governance quality and political economy variables; and (iv) extend the methodology to on-grid solar efficiency to enable a comparative analysis of grid vs. off-grid delivery effectiveness.

References

References

- [1] Acemoglu, D., Aghion, P., Bursztyn, L., & Hemous, D. (2012). The environment and directed technical change. *American Economic Review*, **102**(1), 131–166.
- [2] Banker, R. D., Charnes, A., & Cooper, W. W. (1984). Some models for estimating technical and scale inefficiencies in data envelopment analysis. *Management Science*, **30**(9), 1078–1092.
- [3] Charnes, A., Cooper, W. W., & Rhodes, E. (1978). Measuring the efficiency of decision making units. *European Journal of Operational Research*, **2**(6), 429–444.
- [4] IRENA. (2020). *Renewable Power Generation Costs in 2020*. International Renewable Energy Agency.
- [5] Mardani, A., Zavadskas, E. K., Streimikiene, D., Jusoh, A., & Khoshnoudi, M. (2017). A comprehensive review of data envelopment analysis (DEA) approach in energy efficiency. *Renewable and Sustainable Energy Reviews*, **70**, 1298–1322.
- [6] Ministry of New and Renewable Energy (MNRE). (2016–24). *Annual Report*. Government of India.
- [7] Simar, L., & Wilson, P. W. (1998). Sensitivity analysis of efficiency scores: How to bootstrap in nonparametric frontier models. *Management Science*, **44**(1), 49–61.
- [8] Simar, L., & Wilson, P. W. (2007). Estimation and inference in two-stage, semi-parametric models of production processes. *Journal of Econometrics*, **136**(1), 31–64.
- [9] Singh, N. (2020). Energy efficiency and economic growth in India: A state-level panel data analysis. *Energy Economics*, **90**, 104859.
- [10] Sueyoshi, T., & Goto, M. (2014). DEA environmental assessment in a time horizon: Malmquist index on fuel mix, electricity, and CO₂ of industrial nations. *Energy Economics*, **40**, 370–382.
- [11] Tulkens, H., & Vanden Eeckaut, P. (1995). Non-frontier measures of efficiency, progress and regress for time series data. *International Journal of Production Economics*, **39**(1–2), 83–97.